October 20, 2026

The Editor-in-Chief
Decision Support Systems

Dear Editor,

We submit the manuscript "Explainable Inventory Decision Support through Computational Intelligence: An Enterprise Framework for FMCG Inventory Management" for consideration as a research article in *Decision Support Systems*.

The paper addresses a recurring decision of distributors of fast-moving consumer goods: setting order quantities and safety stocks for many products that share a procurement budget and warehouse space under uncertain demand and lead times. Following design science research, we design, implement and evaluate a decision support artifact in which feasibility, a certified bound on solution quality and traceable, rule-based explanations are first-class outputs. The artifact couples a joint inventory model, shown to be convex within each price regime, with a Lagrangian lower bound, an adapted Artificial Bee Colony algorithm and an explanation layer, and it is instantiated in a working decision support system with a decision-recording workflow.

We believe the paper fits the journal because its contribution is a reusable design for optimisation-based decision support rather than a new algorithm, and because it reports its evidence with the transparency this design requires: equal-budget comparisons with adapted and generic competitors, certified and exact references, non-parametric statistics with effect sizes, out-of-sample replay, and the conditions under which the proposed method was not the best. The evaluation is synthetic and technical; we state this in the abstract and specify the user evaluation that would test the artifact's effect on decisions.

The manuscript has not been published and is not under consideration elsewhere. Both authors approved the submission. The code and data generator will be made available as stated in the manuscript. The use of an AI assistant is declared as required by the journal's policy.

Yours sincerely,

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